

## ECE 3317

### Homework #7

This assignment is not to be turned in. These are practice problems to help you understand the topics that were covered in Notes 18 (“Reflection and Transmission of Plane Waves”).

- 1) A plane wave in free space has an electric field given by

$$\underline{E} = (4\hat{x} - 4\hat{y} - \sqrt{0.8}\hat{z})e^{-j(k_x x + k_y y + k_z z)}, \text{ where } k_x = 0.4k_0 \text{ and } k_y = 0.2k_0.$$

Determine  $k_z$  and the  $\underline{k}$  vector, and verify that the electric field is perpendicular to the  $\underline{k}$  vector. Then determine the magnetic field vector in the phasor domain from Maxwell's equations, and verify that it is also perpendicular to the  $\underline{k}$  vector.

- 2) Calculate the critical angle for a plane wave going from:

(a) Water ( $\epsilon_r = 81, \mu_r = 1$ ) to air.

(b) Silicon ( $\epsilon_r = 11.68, \mu_r = 1$ ) to air

- 3) White light is composed of the entire visible spectrum. The index of refraction  $n$  for most materials is a weak function of wavelength  $\lambda$ , often described over a limited frequency region by Cauchy's equation,  $n = A + B/\lambda^2$ . A beam of white light is incident at  $35^\circ$  on a piece of glass with  $A = 1.4$  and  $B = 6 \times 10^{-15} \text{ m}^2$ . What are the transmitted angles for colors violet (400 [nm]), blue (450 [nm]), green (550 [nm]), yellow (600 [nm]), orange (650 [nm]), and red (700 [nm])? This separation of colors is the dispersive effect of a prism.

- 4) A microwave signal at 2.0 GHz from a cell-phone base station bounces off of a building, which is modeled as a lossless nonmagnetic material having a relative permittivity of  $\epsilon_r = 4$ . Assume that the angle of incidence is  $45^\circ$ . What is the percent power reflected if the wave is a  $\text{TM}_z$  polarized wave? What is the percent power reflected if it is a  $\text{TE}_z$  polarized wave? What is the percent power reflected if it has equal powers in both polarizations?

- 1) A plane wave in free space has an electric field given by

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$$a) \quad k_x^2 + k_y^2 + k_z^2 = k_0^2 \quad k_z = \sqrt{k_0^2 - k_x^2 - k_y^2} = \sqrt{k_0^2 - (0.4k_0)^2 - (0.2k_0)^2} = \sqrt{0.8} \quad k_0 = 0.8944k_0$$

$$b) \quad \underline{k} = k_0 (0.4\hat{x} + 0.2\hat{y} + 0.8944\hat{z})$$

$$c) \text{ Find } \underline{H}: \quad \underline{H} = \frac{\underline{k} \times \underline{E}}{\omega\mu} = \frac{k_0}{\omega\mu} \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0.4 & 0.2 & 0.8944 \\ 4 & -4 & -\sqrt{0.8} \end{vmatrix} e^{-jk_r r}$$

$$\underline{H} = \frac{k_0}{\omega\mu} e^{-jk_r r} (3.339\hat{x} + 3.935\hat{y} - 2.4\hat{z})$$

- 2) Calculate the critical angle for a plane wave going from:

(a) Water ( $\epsilon_r = 81, \mu_r = 1$ ) to air.

(b) Silicon ( $\epsilon_r = 11.68, \mu_r = 1$ ) to air

$$a) \quad N_{\text{Water}} \sin \theta_c = N_{\text{air}} \sin \theta_t$$

$$N_{\text{Water}} = \sqrt{81 \cdot 1} = 9$$

$$N_{\text{air}} = \sqrt{1} = 1 \quad \theta_t = 90^\circ$$

$$\theta_c = \sin^{-1} \left( \frac{\sin 90^\circ}{9} \right) = 6.38^\circ$$

$$b) \quad N_{\text{silicon}} \sin \theta_c = N_{\text{air}} \sin \theta_t$$

$$N_{\text{silicon}} = \sqrt{11.68(1)}$$

$$N_{\text{air}} = \sqrt{1} = 1$$

$$\theta_c = \sin^{-1} \left( \frac{\sin 90^\circ}{\sqrt{11.68}} \right) = 17.014^\circ$$

- 3) White light is composed of the entire visible spectrum. The index of refraction  $n$  for most materials is a weak function of wavelength  $\lambda$ , often described over a limited frequency region by Cauchy's equation,  $n = A + B/\lambda^2$ . A beam of white light is incident at  $35^\circ$  on a piece of glass with  $A = 1.4$  and  $B = 6 \times 10^{-15} \text{ m}^2$ . What are the transmitted angles for colors violet (400 [nm]), blue (450 [nm]), green (550 [nm]), yellow (600 [nm]), orange (650 [nm]), and red (700 [nm])? This separation of colors is the dispersive effect of a prism.

$$\text{red} = \sin^{-1} \left( \frac{\sin 35^\circ}{\left(1.4 + \frac{6E-15}{700E-9}\right)^2} \right) = 23.96^\circ$$

$$\text{orange} = \sin^{-1} \left( \frac{\sin 35^\circ}{\left(1.4 + \frac{6E-15}{650E-9}\right)^2} \right) = 23.93^\circ$$

$$\text{yellow} = \sin^{-1} \left( \frac{\sin 35^\circ}{\left(1.4 + \frac{6E-15}{600E-9}\right)^2} \right) = 23.88^\circ$$

$$\text{green} = \sin^{-1} \left( \frac{\sin 35^\circ}{\left(1.4 + \frac{6E-15}{550E-9}\right)^2} \right) = 23.83^\circ$$

$$\text{blue} = \sin^{-1} \left( \frac{\sin 35^\circ}{\left(1.4 + \frac{6E-15}{450E-9}\right)^2} \right) = 23.65^\circ$$

$$\text{Violet} = \sin^{-1} \left( \frac{\sin 35^\circ}{\left(1.4 + \frac{6E-15}{400E-9}\right)^2} \right) = 23.52^\circ$$

- 4) A microwave signal at 2.0 GHz from a cell-phone base station bounces off of a building, which is modeled as a lossless nonmagnetic material having a relative permittivity of  $\epsilon_r = 4$ . Assume that the angle of incidence is  $45^\circ$ . What is the percent power reflected if the wave is a  $TM_z$  polarized wave? What is the percent power reflected if it is a  $TE_z$  polarized wave? What is the percent power reflected if it has equal powers in both polarizations?

Given:  $f = 2.0 \text{ E6 [Hz]}$ ,  $\epsilon_r = 4$ ,  $\mu_r = 1$  (lossless, nonmagnetic)

$$\theta_i = 45^\circ$$

Medium 1: air ( $\epsilon_1 = \epsilon_0$ ,  $\eta_1 = \eta_0$ )

Medium 2: building ( $\epsilon_2 = 4\epsilon_0$ )

$$\Gamma = \frac{Z_{TM2} - Z_{TM1}}{Z_{TM2} + Z_{TM1}}$$

Find  $Z_{TM1}$ :

$$Z_{TM1} = \frac{K_{z1}}{W\epsilon_1} = \frac{W\sqrt{\epsilon_0} \cos \theta_i}{W\epsilon_0} = \sqrt{\frac{1}{\epsilon_0}} \cos \theta_i = \sqrt{\frac{\mu_0}{\epsilon_0}} \cos \theta_i = \eta_0 \cos \theta_i = 177 [\Omega]$$

First find  $\theta_t$  from Snell's law:

$$k_1 \sin \theta_i = k_2 \sin \theta_t \Rightarrow W\sqrt{\epsilon_0} \sin 45^\circ = W\sqrt{\epsilon_0 \epsilon_r} \sin \theta_t$$

$$\sin \theta_t = \frac{\sin 45^\circ}{\sqrt{4}} = \frac{\sin 45^\circ}{2}$$

$$\theta_t = \sin^{-1} \left( \frac{\sin 45^\circ}{2} \right) = 20.7^\circ$$

Finding  $Z_{TM2}$ :

$$Z_{TM2} = \frac{K_{z2}}{W\epsilon_2} = \frac{W\sqrt{\epsilon_0 \epsilon_r} \cos \theta_t}{W\epsilon_0 \epsilon_r} = \sqrt{\frac{\mu_0}{\epsilon_0 \epsilon_r}} \cos \theta_t = \frac{\eta_0}{\sqrt{\epsilon_r}} \cos \theta_t = \frac{\eta_0}{2} \cos (20.7^\circ) = 176 [\Omega]$$

$$\Gamma = \frac{176 - 177}{176 + 177} = 0.2038$$

$$\% \text{ Power reflected} = (100)(0.2038)^2 = 4.15\%$$

b)  $TE_z$  Polarization

$$\Gamma = \frac{Z_{TE2} - Z_{TE1}}{Z_{TE2} + Z_{TE1}}$$

$$\text{Finding } Z_{TE1}: \frac{W\mu_0}{K_{z1}} = \frac{W\mu_0}{W\sqrt{\epsilon_0} \cos \theta_i} = \sqrt{\frac{\mu_0}{\epsilon_0}} \frac{1}{\cos \theta_i} = \frac{\eta_0}{\cos \theta_i} = \frac{\eta_0}{\cos 45^\circ} = 534 [\Omega]$$

$$\text{Finding } Z_{TE2}: \frac{W\mu_0}{K_{z2}} = \frac{W\mu_0}{W\sqrt{\epsilon_r \epsilon_0} \cos \theta_t} = \frac{\eta_0}{\sqrt{\epsilon_r} \cos \theta_t} = \frac{\eta_0}{2 \cos (20.7^\circ)} = 202 [\Omega]$$

$$\Gamma = \frac{202 - 534}{202 + 534} = -0.4514$$

$$\% \text{ Power Reflected} = (100)(0.4514)^2 = 20.38\%$$

c.) Equal Powers in Both Polarizations

Assume incident power density =  $1 \text{ [W/m}^2\text{]}$ , So each Polarization carries  $0.5 \text{ [W/m}^2\text{]}$

$$\% \text{ power reflected} = \left( \frac{0.0415 + 0.2038}{2} \right) (100) = (24.53\%) \left( \frac{1}{2} \right)$$

$$\% \text{ power reflected} = 12.265\%$$